choice of splitting strategy (given by the definition of the splitting pieces) along with a prescribed sequence of updates defines a splitting method.

There are a great number of plausible splitting strategies for Langevin dynamics. For example we may split the equations into purely deterministic and stochastic parts, break the stochastic terms into dissipative and random components, or choose to decompose the vector field into N_c one-dimensional problems that can be solved to high-accuracy. For some potential energy functions, it may be beneficial to decompose the potential into many terms $U(\boldsymbol{q}) = U_1(\boldsymbol{q}) + U_2(\boldsymbol{q}) + \ldots$, and use each term in a separate splitting piece in order to give a scheme improved stability or computational properties. In any case, it is not obvious *a priori* which splitting strategies will have more favorable sampling properties.

As propagating the Ornstein-Uhlenbeck process exactly will preserve the canonical distribution ρ_β, it is reasonable to assume that this is a good choice for one of the building blocks in a general splitting scheme. As the exact solution of the OU process is defined for any value of $\gamma \geq 0$, we may hope that our discretization scheme will be tolerant for extremely large frictions, although there is no guarantee that the resulting discretization remains consistent in the limiting case.

Other than the Ornstein-Uhlenbeck process, the remaining piece of the splitting corresponds to Newtonian constant-energy (microcanonical) Hamiltonian dynamics. Since Hamiltonian dynamics leaves invariant any function of the energy, its corresponding Fokker-Planck operator (in this case the Liouvillian, $\mathcal{L}_H^\dagger = -\mathcal{L}_H$) will preserve distributions that are functions of the Hamiltonian H. This implies in particular that it preserves ρ_β, which is proportional to $\exp(-\beta H)$. Thus the forward propagator associated to the Hamiltonian system automatically preserves the Gibbs distribution, and we have

$$e^{t\mathcal{L}_H^\dagger}\rho_\beta = \rho_\beta.$$

Formally, we could combine the flow map $\mathscr{F}_h^{\mathrm{H}}$ of Hamiltonian dynamics with an exact solve of the OU process to create a sampling iteration which preserves ρ_β, as we also have

$$e^{t\mathcal{L}_O^\dagger}\rho_\beta = \rho_\beta.$$

To make this practical in the general case, where the Hamiltonian system is not integrable (and we cannot explicitly compute its flow map), we need to be able to compute (or approximate) the maps involved.

This suggests to further split the Hamiltonian system, as we have done in deriving symplectic integrators in Chaps. 2 and 3.

We therefore consider dividing the Langevin system (7.4) into parts

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}\\ \boldsymbol{p}\end{bmatrix} = \underbrace{\begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{p}\\ \mathbf{0}\end{bmatrix}\mathrm{d}t}_{\mathrm{A}} + \underbrace{\begin{bmatrix}\mathbf{0}\\ -\nabla U(\boldsymbol{q})\end{bmatrix}\mathrm{d}t}_{\mathrm{B}} + \underbrace{\begin{bmatrix}\mathbf{0}\\ -\gamma\boldsymbol{p}\mathrm{d}t + \sigma\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}\end{bmatrix}}_{\mathrm{O}}, \tag{7.5}$$

which we label A, B and O, where each of the three parts may be solved "exactly", with individual updates given as

$$\begin{aligned}\mathscr{U}_h^{\mathrm{A}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{p}, \boldsymbol{p}),\\ \mathscr{U}_h^{\mathrm{B}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q}, \boldsymbol{p} - h\nabla U(\boldsymbol{q})),\\ \mathscr{U}_h^{\mathrm{O}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q}, e^{-\gamma h}\boldsymbol{p} + \sqrt{\mathrm{k_B T}\,(1 - e^{-2\gamma h})}\boldsymbol{M}^{1/2}\mathbf{R}),\end{aligned}$$

where $\mathbf{R}$ is a vector of i.i.d. normal random numbers. For the deterministic pieces A and B, the updates correspond to their flow maps, while the update for the O piece corresponds to the solution derived in (7.3). The relevant forward propagators for each piece of the Hamiltonian part are computed through the adjoint of the Lie derivative as

$$\mathcal{L}_{\mathrm{A}}^{\dagger} = -\boldsymbol{M}^{-1}\boldsymbol{p}\cdot\nabla_q, \qquad \mathcal{L}_{\mathrm{B}}^{\dagger} = \nabla U(\boldsymbol{q})\cdot\nabla_p.$$

Using this splitting strategy, we can define a family of schemes by the desired sequence of updates, codified as a string using the alphabet "ABO". For example, the method denoted $[\![\mathrm{ABO}]\!]$ (acting from left to right) has a resulting update

$$\mathscr{U}_h^{[\![\mathrm{ABO}]\!]} = \mathscr{U}_h^{\mathrm{O}} \circ \mathscr{U}_h^{\mathrm{B}} \circ \mathscr{U}_h^{\mathrm{A}}.$$

This corresponds to using the adjoint symplectic Euler scheme to solve the Newtonian part of the Langevin dynamics SDE, followed by an exact OU solve. An alternative is to use velocity Verlet for the Hamiltonian part, resulting in a scheme denoted $[\![\mathrm{BABO}]\!]$ with associated map

$$\mathscr{U}_h^{[\![\mathrm{BABO}]\!]} = \mathscr{U}_h^{\mathrm{O}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_h^{\mathrm{A}} \circ \mathscr{U}_{h/2}^{\mathrm{B}}.$$

Note that the presence of two B characters in the string results in halving the corresponding piece's time step length. As we need the method to be consistent, we require that the each of the A, B and O pieces are propagated in time for a total time of h. As a result we use the convention that if a letter appears k times in a method's string, then each individual update piece uses a timestep h/k. For example, the scheme denoted $[\![\mathrm{OABOAOBAO}]\!]$ has an update

$$\mathscr{U}_h^{[\![\mathrm{OABOAOBAO}]\!]} = \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/4}^{\mathrm{O}},$$

where the random numbers used in each O update are independent. This scheme will require a total of $4\mathrm{N}_c$ normal random numbers and two force evaluations per step (the force must be recomputed at the first B step following an A step; the number of force evaluations required is not directly related to the length of the string).

A class of methods like those considered above were discussed in [46] and given the general title of "Geometric" Langevin Algorithm (GLA). The idea of a GLA